



# Rachit Mangawa

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## EDUCATION

| Course           | Board / University                     | Year           | CGPA / % |
|------------------|----------------------------------------|----------------|----------|
| B.Tech (CSAI)    | Netaji Subhas University of Technology | 2023 - Present | 7.34     |
| CBSE (Class XII) | SR.D. International School             | 2022 - 2023    | 88.20    |
| CBSE (Class X)   | Swami Keshwanand Convent School        | 2020 - 2021    | 90.40    |

## WORK EXPERIENCE

### • Unique Identification Authority of India (UIDAI) | Software Developer Intern

June 2026 - September 2026

#### Project 1: Offline OCR + LLM Grievance Processing System

- Built fully **offline** grievance-letter pipeline for scanned PDFs in **12 languages**: **OpenCV** preprocessing (Otsu, deskew, CLAHE) → **Tesseract OCR** → **Ollama** (gemma3:4b) → schema-strict **JSON**.
- Optimized for **8GB CPU-only** hardware: disk-streamed uploads, LLM **boundary detection** to split multi-letter bundles, degraded-page flagging, and malformed-output fallback; **1–2 min/letter**.
- Shipped **HTML/JS review** UI with side-by-side scan and summary, **confidence scores**, and urgency/type filters for officer triage.

#### Project 2: Contact Center Operations & Analytics Platform

- Built **Dash + FastAPI + Pandas** platform unifying **4 data sources** (CCF, CDR, UniMate, APR) across **2 vendors**, replacing manual **Excel** reporting; auto-computes **SL, AHT, avg hold** against **85% / 240s / 20s** targets with cross-filtering, heatmaps, agent quadrant, and **180-day** range.
- Designed **async ETL** with **WebSocket** progress and stateless parsers: ingests **50MB / 100K+ row** workbooks in **5–15s** via **1,000-row** batches and **idempotent upserts**; **APScheduler** 3-min auto-ingest on **SQLite WAL**.
- Secured with **JWT RBAC**, **tenant isolation**, **X-Forwarded-For** audit logs, **bcrypt** lazy rehash, 30-min inactivity logout, and 90-day log pruning; architected for **1,000+ users**.

## ACHIEVEMENTS

- Pupil** on [Codeforces](#) (max rating **1341**);
- Knight** on [LeetCode](#) (max rating **1982**, top **2.91%**);
- Global Rank 316** of 38,291 (top **0.83%**) – [LeetCode Biweekly Contest 190](#).
- Open Source**: Debugged a numerical edge case in **TensorFlow**'s autodiff - `tf.math.bessel_i1`'s gradient was hardcoded to return **1.0** at  $x=0$ , a removable singularity where the correct limit is **0.5**; fixed the imputed value in `math_grad.py` and added regression tests in `special_math_ops_test.py` covering `bessel_i1` and `bessel_i1e`. **Merged** - [Issue #120578](#), [PR #120944](#).

## PROJECTS

### • HeartFlow OS | [GitHub](#) | [Live Demo](#)

Feb 2026 – April 2026

TypeScript | Next.js | Python | FastAPI | WebSockets | Scikit-Learn | SHAP | C/C++ | Docker | PyTest

- Built **AST transpiler** compiling **Scikit-Learn** models into zero-dependency, **malloc-free C headers** for **ESP32 / Cortex-M** deployment.
- Engineered **INT8 quantization** engine, cutting microcontroller flash footprint by **~75%**.
- Automated **MLOps** pipeline for soft-voting ensemble with **SHAP** explainability, test-set **F1** gating, and automatic rollback on regression.
- Built **FastAPI** backend streaming real-time **WebSocket** telemetry to fault-tolerant **Next.js** dashboard with auto-reconnect.
- Verified C++/Python model-output parity with cross-language **PyTest** suite; containerized full stack with **Docker Compose**.

### • Semantic Comment Analyzer: Intent-Driven NLP Platform | [GitHub](#)

Oct 2025 – Dec 2025

Python 3.13 | FastAPI | Sentence-Transformers | Scikit-Learn | Hugging Face Transformers | RoBERTa | Pandas | PyTest

- Built **FastAPI** NLP service (POST /api/analyze) returning **intent, confidence, sentiment, entities, urgency, tone, recommended action** as one JSON.
- Replaced slow zero-shot classification with **all-MiniLM-L6-v2** embeddings + **Logistic Regression / SVM** head, serving **thousands of requests/sec** with **dynamic intent labels**.
- Built **occlusion explainability**: masks each word, re-runs classifier, and measures confidence delta for **word-level intent attribution**.
- Integrated **sentiment** (HF pipeline, **RoBERTa** fine-tune script), **NER** (ORG, LOC, PER), and rule-based **business-heuristics** layer for urgency, tone, and routing.
- Runs **fully local** (training + inference, no cloud) on **4GB+ RAM**; Pandas CSV batch handler and **PyTest** suite.

## SKILLS

- Languages**: C++, Python, TypeScript, SQL, Bash
- Backend**: FastAPI, REST APIs, WebSockets, JWT/RBAC, SQLite, APScheduler, Next.js
- AI / ML**: PyTorch, Hugging Face Transformers, Sentence-Transformers, Scikit-Learn, SHAP, OpenCV, Tesseract, Ollama, RAG
- Data / Visualization**: Pandas, NumPy, Plotly Dash
- Developer Tools**: Git, GitHub Actions, Docker, PyTest, Linux

## RELATED COURSEWORK

Machine Learning, Artificial Intelligence (AI), Computer Vision, Natural Language Processing (NLP), Operating Systems (OS), Computer Networks, Data Structures, Database Management System (DBMS)